STUMPRIDER · PROJECT PLAN

StumpRider project plan

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25 (SMR-DEC-001 O6 to O9) written in; O10 back with Amish
0.1	2026-10-24	Amish Chadha	First issue in the kit 1.9.1 upgrade; stage P3 reached on paper, P4 on hold under the portfolio cap

StumpRider project plan

StumpRider is at TRL 3 and stage P3 (design outputs): model at D3 for three concept options, drawings, priced BOM, build plan, decisions register, design FMEA and traceability for option A. The next gate, P4 (verification at TRL 4), is on hold under the portfolio caps of TRL 3 and P3. Decisions requested from Amish: the concept selection (O7, option A recommended) and the heel limit for a sudden jerk (O6, option A recommended). Method: STANDARDS sections 19 and 23.

> **Safety:** Every planned trial below is from a moored canoe or a bench; nobody enters the water, every person wears a life jacket, and no kit is used on a snag before its pin reel is calibrated and the heel test is passed.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Design and calculations	Design lead	Model (D3), STEP, drawings, SMR-CAL-001, SMR-STM-001	Done to P3; S7 plausible after O6 (2026-10-25); S3b open with Amish (O10)
Concept selection	Amish Chadha	SMR-OPT-001, SMR-DDR-005	Option A chosen by Amish, 2026-10-25 (O7)
Requirements and traceability	Design lead	SMR-URS-001, SMR-PRS-001, SMR-RTM-001	Done to P3; targets to confirm with a partner
Partner	Project lead	A child-protection NGO and a landing-site committee as co-design partners	Not started (risk P1)
Regulatory	Partner liaison	Answers to the "to assess" items of SMR-REG-001	Not started
Verification (P4, TRL 4)	Test partner	Pin calibration, bench snags, moored heel tests by canoe class, staged-snag trials, build trial, six-month wet storage	On hold (portfolio cap)
Validation (P5)	Partner	Use by crews on the lake with the protection programme	On hold

Resources needed

Table 2. Resources (P4 planned work, on hold; costs are estimates from the BOM, not quotes).

Resource	Deliverable	Effort	Estimated cost (USD)
Village smith	Three aluminium kits and one bamboo pole from SMR-BLD-001	About 4 working days	Parts USD 299.50 (SMR-CAL-001 [C2]); labour to assess
CalRig or a spring balance rig	Ten pins per reel broken (PR-011)	1 day	Within the prototype run
Moored canoe, three crew in life jackets, inclinometer	Static and jerk heel (PR-003, O6)	1 day per canoe class	To assess with the partner
University fisheries department	Staged-snag trials (PR-001, PR-004, PR-005)	5 days	To assess
Partner crews	Six-month wet storage (PR-008)	6 months	To assess

The value-engineering target is USD 2,000 (`project.yaml` budget_usd, unchanged); the prototype run is estimated at USD 299.50.

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB with users and stakeholders, STM, OPT, REG, FEA, PLN	Met (2026-10-24)
P1 Definition	URS, PRS, RSK with hazard analysis, MKT	Met (2026-10-24)
P2 Design inputs	PRS with targets and verification, CAL, RTM, concept selection DDR, STEM check closed on paper	Met on paper; concept selection decided 2026-10-25; S3b (O10) awaiting Amish
P3 Design outputs	Model, STEP, drawings, priced BOM, BLD, DEC, design FMEA, traceability complete	Met (2026-10-24)
P4 Verification	Test plans and reports, verification report	On hold (cap)
P5 Validation	Field trials with a partner	On hold (cap)
P6 Commercialization	Only with a named deployment partner; for StumpRider a distribution plan, not a sale	On hold (cap)

Next steps and decision requested

1. Amish: decide O10 again (hook head tip leg: accept the case, 14 mm bar, or a gusset to the flat, each trading PR-012 against PR-007). O6 to O9 were decided on 2026-10-25.
2. Approach the first co-design candidates (Challenging Heights; a Volta landing-site committee through the Fisheries Commission; a university fisheries department) once Amish agrees.
3. When the portfolio cap lifts: build to SMR-BLD-001 and run the first checks.

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